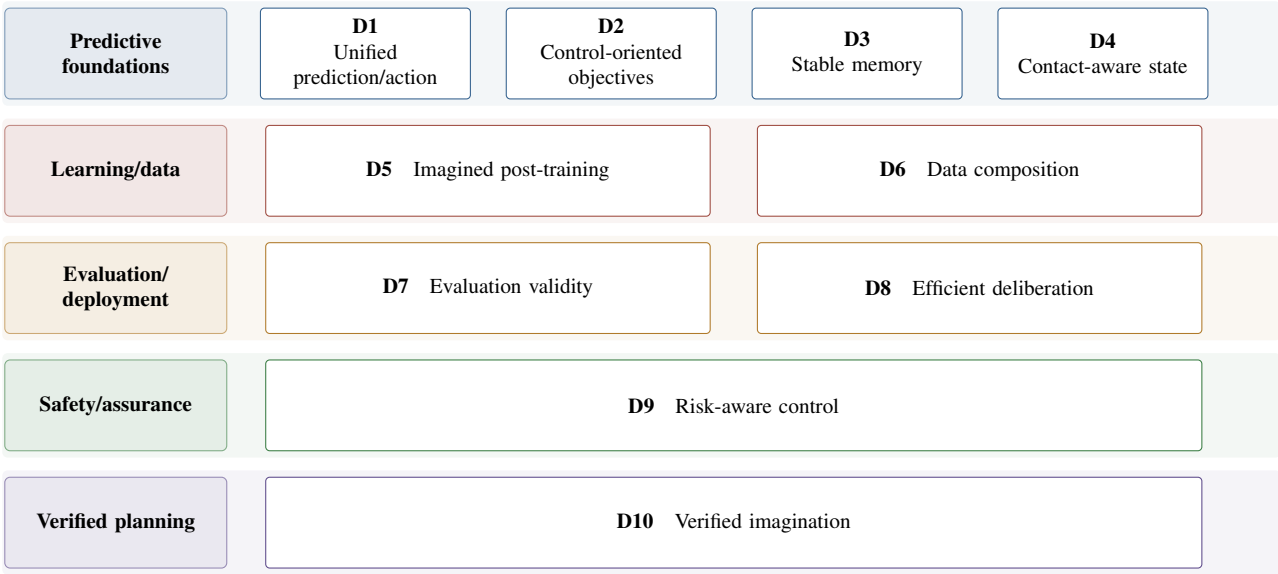

Research directions grouped by theme; numbering matches Sec. VIII and does not imply maturity or a deployment schedule.